

# DURGESH UNDE

## AI Engineer & Founder | LLM Fine-Tuning, AI Agents & Automation

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### PROFESSIONAL SUMMARY

AI engineer and founder building LLM-powered software, AI agents, automation, and conversational AI systems. Hands-on work spans LLM fine-tuning, model evaluation, grounding, alignment dataset curation, and human-in-the-loop QA. Built XYVERION 3.0 with Python/PyTorch, 4-bit QLoRA/PEFT, Win32 automation, Edge WebView2, SQLite, and local model infrastructure.

### TECHNICAL SKILLS

**LLM & AI:** LLM Fine-Tuning; 4-bit QLoRA; PEFT; Transformers; AI Evaluation; Alignment Dataset Curation; Data Annotation; Grounding; Human-in-the-Loop QA; Conversational AI; AI Agents; Tool Calling; Automation

**Programming & ML:** Python; PyTorch; JavaScript (ES6+); HTML5; CSS3

**Systems & Data:** Win32 API; OS Automation; SQLite; Microsoft Edge WebView2; Weight Fusion; Safetensors; Git; GPU VRAM Management

### EXPERIENCE

#### Founder & AI Engineer | XYVERION AI

Sep 2026 - Present

- Build AI software spanning conversational AI, agentic workflows, desktop automation, LLM-based systems, and practical product applications.
- Develop model-integrated workflows covering tool calling, local model execution, application interfaces, automation, and deployment.
- Build and iterate AI systems across model integration, application interfaces, automation components, and software architecture.

#### AI Data Annotation & Evaluation Contractor | Innodata India Pvt. Ltd.

August 2026 - Present

- Perform data annotation, content moderation, and quality evaluation for AI/ML training projects, including grounding and reference-expression workflows.
- Execute dense-structured grounding, CUA-general grounding, and high-resolution reference-expression annotation; review outputs for accuracy, relevance, consistency, and quality.
- Conduct linguistic and semantic review while maintaining annotation quality, AHT, benchmarks, QA requirements, and delivery timelines.

#### AI Training & Data Annotation Contributor | Outlier AI

2026 - July 2026

- Contributed to AI training and model evaluation through structured human feedback, image annotation, data labeling, and content-quality assessment.
- Collected multilingual speech and voice data in English, Hindi, and Marathi for AI training workflows.

### SELECTED AI PROJECT

#### XYVERION 3.0 - Autonomous Desktop AI & Local SLM Engine | Python / PyTorch / QLoRA / PEFT / Transformers / Win32 / SQLite / Edge WebView2

- Curated a 1,749-example multi-turn instruction dataset covering deductive reasoning, tool-calling schemas, and creator-attribution guardrails.
- Fine-tuned Qwen 1.5B on an NVIDIA GTX 1650 4GB using 4-bit QLoRA (NF4, r=16, alpha=32), reducing validation loss from 1.6048 to 1.2473 over 85 optimization steps.
- Fused 112 LoRA adapter shards into base-model weights after training, removing runtime adapter overhead from the resulting model package.
- Implemented a desktop AI stack with Win32 voice/OS control (55+ commands), Edge WebView2 UI, SQLite episodic memory, and GPU VRAM management.

### RESEARCH

- Fake News Detection Using Machine Learning Techniques
- Enhancing Cybersecurity with Machine Learning-Based Threat Detection
- AI in Healthcare Diagnostics: Beyond Human Accuracy
- AI in Climate Change Prediction and Environmental Sustainability
- The Evolution of Generative AI Models: From Text to General Intelligence

### EDUCATION

**Higher Secondary Certificate (HSC) - Science** 78.50% | 2020 - 2022

**Secondary School Certificate (SSC)** 85.20% | 2020

### CERTIFICATION

**IELTS Academic - Overall Band 7.5 (CEFR C1)** | Sep 2025 | IDP

### LANGUAGES

English - Advanced | Hindi - Advanced | Marathi - Advanced